

Pharmacy Database Management System

Course: CS210 Final Project

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GitHub: <https://github.com/nav73-commits/CS210-Final-Project>

1. Project Overview

The Pharmacy Database Management System is designed to handle essential pharmacy operations such as data entry, retrieval, inventory tracking, sales management, and customer records. The system provides a centralized database-driven solution to improve accuracy in medicine dispensing and streamline daily operations.

2. Entity-Relationship (ER) Diagram

The database is designed in the Third Normal Form (3NF) to ensure data integrity and reduce redundancy.

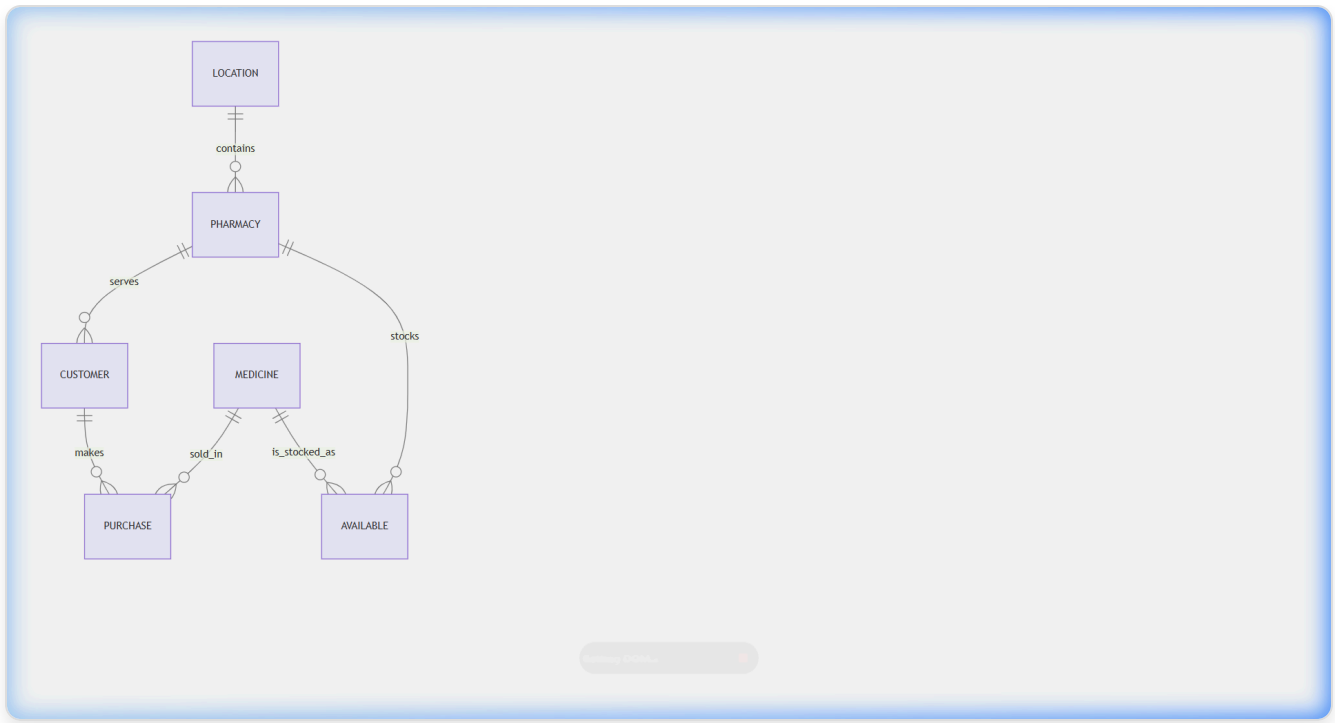


Figure 1: Relational Database Schema (3NF)

3. System Implementation

The project consists of a Python-based Flask application and a Jupyter Notebook for data analysis. The two components are synchronized through a shared SQLite database.

3.1 Administrative Dashboard

The web dashboard provides real-time metrics, including total revenue, transaction counts, and low-stock alerts.

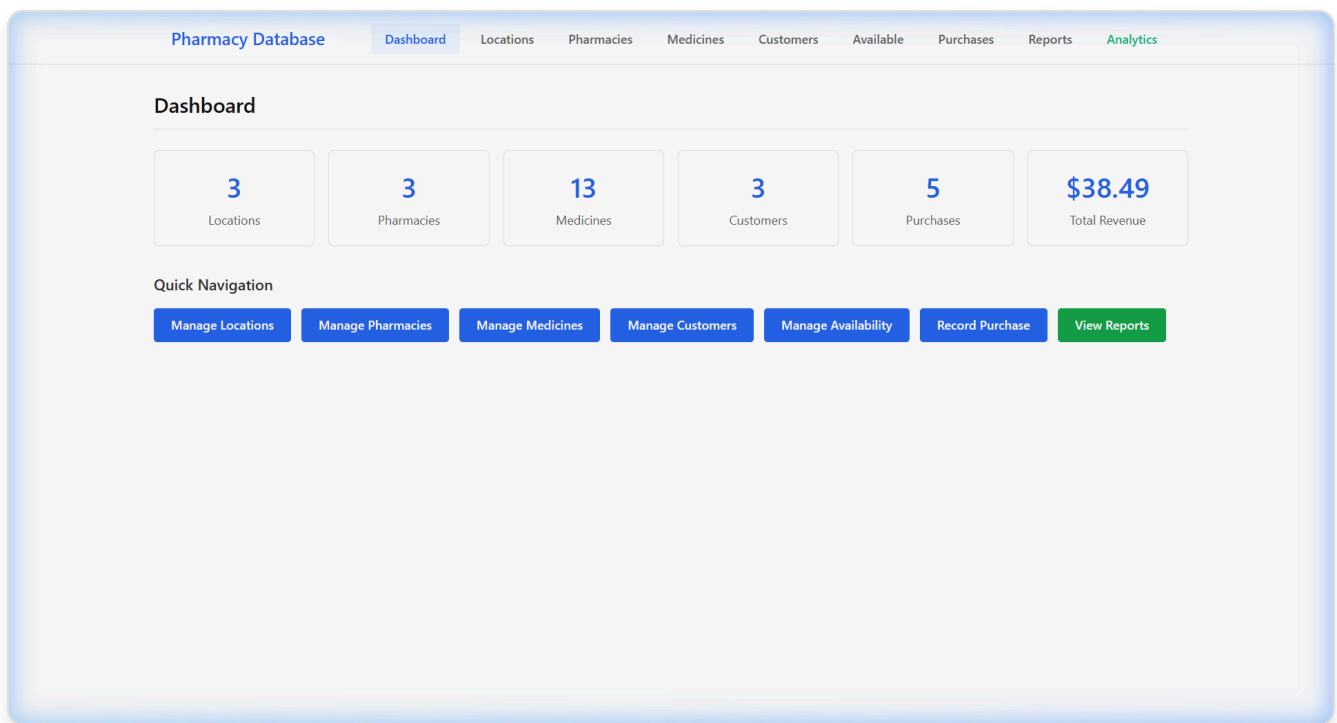


Figure 2: Web Frontend Dashboard

3.2 Advanced Analytics

The analytics module provides deep insights into pharmacy performance, medicine demand, and customer spending patterns.

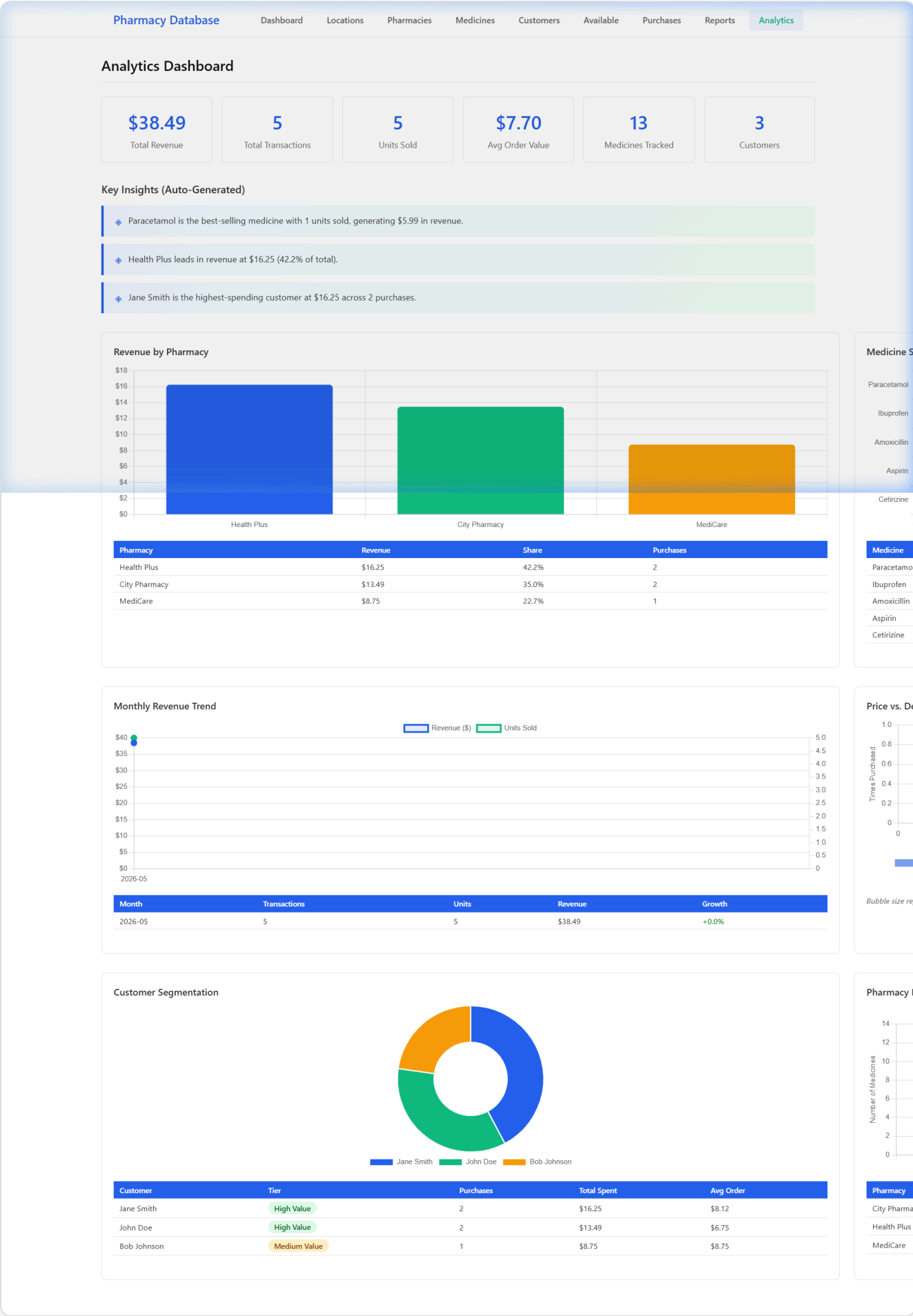


Figure 3: Analytics and Revenue Reporting

3.3 Inventory Management

Users can manage medicine records, update costs, and track dosage information through a user-friendly interface.

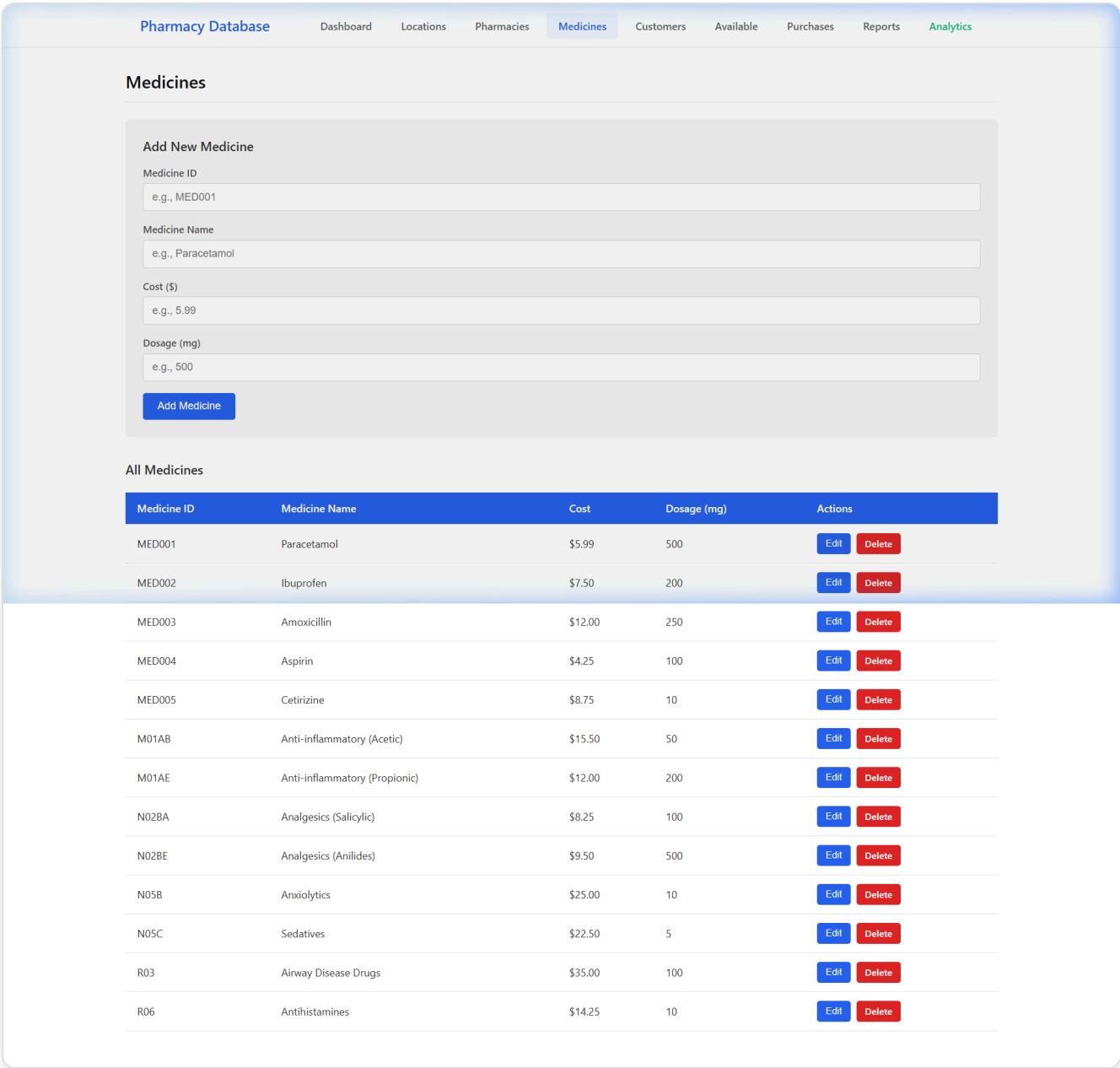


Figure 4: Medicine Inventory Management

4. Conclusion

This project successfully implements a comprehensive pharmacy management solution that addresses real-world operational challenges through structured database design and modern

web technology.